

Product Differentiation & Demand Estimation

Foreshadowing Uses of Demand Estimation

- What would be the effect on consumer surplus if:
 - Two firms merged?
 - A new product is introduced?
 - Advertising is banned?
 - A policy reduces the costs of switching health insurance or retirement plans?
- By making a few (hopefully reasonable assumptions), can calculate **dollar estimates of effects of magnitudes** rather than “just consumer surplus will fall”₂

Broad Questions

- Motivation
 - Products are clearly differentiated
 - Consumers have heterogeneous preferences
- Good teaching works:
 - Train's book Discrete Choice Methods with Simulation
 - Nevo's "Empirical Models of Consumer Behavior"
 - **Nevo's "Practitioners Guide"**

The Dimensionality Problem

- In general, $O(J^2)$ parameters

- Solutions:

1. Impose symmetry across products

- E.g. CES demand

$$U(q_1, \dots, q_J) = \left(\sum_{i=1}^J q_i^\rho \right)^{1/\rho}$$

2. Collapse product into segments and allow flexible substitution within group g

- Multi-stage budgeting (see Deaton & Muellbauer)
- E.g. AIDS within-group expenditure eqn

$$s_{jt} = \alpha_j + \beta_j \ln(y_{gt}/\pi_{gt}) + \sum_{k=1}^{J_g} \gamma_{jk} \ln(p_{kt}) + \varepsilon_{jt}$$

3. Collapse products to bundles of (a relatively small number of) **characteristics...**

Benefits of Viewing a Good as a Bundle of Characteristics

- Reduces dimensionality problem so fewer than $O(J^2)$ parameters
- Allows us to do counterfactual experiments where the product characteristics change
 - Prices → effects of mergers
 - Other characteristics → value of new goods

Discrete Choice Estimation

- We'll discuss more broadly next, but generally:
 - **Maximum likelihood** feasible in some settings
 - If worried that price is correlated with unobserved product characteristics (“endogeneity”), **GMM** usually used